

RAIN WATER HARVESTING SYSTEMS FOR URBAN AREAS

(With Approximate Cost)

This document provides standard designs of Rain Water Harvesting (RWH) systems commonly used in urban areas along with indicative costs. The designs vary based on plot size, water requirement, and recharge/storage method.

1. Simple Recharge Pit for Individual Houses

- Recharge pit with filter media (sand, gravel, boulders)
- Typical size: ~1.5 m diameter, ~2 m depth
- Approximate Cost: Rs. 45,000

2. Recharge Trench

- Used in larger residential plots or group housing
- Brick masonry with filter media
- Approximate Cost: Rs. 1,05,000

3. Recharge Shaft with Bore

- Suitable for deep aquifers
- Includes slotted PVC pipe up to 30 m depth
- Approximate Cost: Rs. 1,40,000

4. Small Filter Pit

- For small rooftop areas
- Square pit with sand and gravel layers
- Approximate Cost: Rs. 30,000

5. Mini Filter Pit

- Compact design for limited space
- Approximate Cost: Rs. 10,000

6. Circular Bore Recharge Pit

- Circular pit connected to bore well
- Approximate Cost: Rs. 10,000

7. Storage cum Recharge Tank (Small)

- Storage tank: 2.75 x 2 x 1.9 m
- Filter chamber included
- Approximate Cost: Rs. 2,60,000

8. Storage cum Recharge Tank (Large)

- Storage tank: 4 x 3 x 2.25 m
- Two filter chambers
- Approximate Cost: Rs. 7,00,000

9. Filter Chamber with Recharge Well

- Brick filter chamber connected to recharge well
- Approximate Cost: Rs. 1,65,000

10. Surface Filter Chamber

- Nylon cloth, sand, brick ballast layers
- Approximate Cost: Rs. 75,000

11. Deep Recharge Well with Filter

- Flexible conductor pipe and recharge well
- Approximate Cost: Rs. 1,80,000

12. Large Recharge Structure (Community Use)

- Large masonry recharge system
- Approximate Cost: Rs. 3,60,000

13. Dry Well System

- Roof water diverted to dry well through filter
- Approximate Cost: Rs. 85,000

14. Recharge Pit in Park / Open Area

- Used for groundwater recharge in public parks
- Approximate Cost: Rs. 60,000 per pit

15. Surface Storage Systems

(A) Individual Houses:

- Simple rooftop rainwater harvesting arrangement
- Approximate Cost: Rs. 40,000 for 200 sq.m roof area

(B) Group Housing:

- Underground storage tank
- Approximate Cost: Rs. 1,00,000 for 500 sq.m roof area

Note:

- Costs are indicative and may vary with site conditions.

- Filter media composition can be modified as per requirement.
- Designs are not to scale and subject to technical approval.

This is a cleaned, text-only version prepared from scanned drawings for documentation and proposal use.